World Indicators Analysis: Health, Economy, Technology and Environment (2016–2024)

REPORT

Data Analysis Project

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Tools Used: Python, Power Bi

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1. Summary

- I built an end-to-end pipeline from the World Bank API through Python (pandas) to a Power BI dashboard. It covers 26 indicators in 7 themes for 217 countries, plus 49 regional and income-group aggregates, for 2016–2024.
- **Income is the master variable.** Log GDP per capita correlates 0.87 (Spearman) with life expectancy and 0.88 with internet use.
- **Health spending as a share of GDP explains almost nothing once income is known.** R^2 stays at 0.70 with or without it.
- **The pandemic left a clear mark.** Average GDP growth was −5.1% in 2020, and life expectancy fell in 84% of countries between 2019 and 2021.
- **Digital catch-up is real.** Lower-middle-income countries gained 26 points of internet use in 2016–2022, against 9 for high-income countries.
- **Data quality changes conclusions.** 2024 is nearly empty, poverty data fills only 20% of country-years, and including aggregates inflated the dashboard's average trade value 6.7×.

2. Objectives

- Build a repeatable, API-driven pipeline for global development indicators.
- Present economic, health, technology, labour, trade, poverty and environment indicators in one interactive dashboard.
- Find relationships between income, health and technology.
- Audit the dashboard for errors and document corrections.

3. Data and Method

3.1 Source and scope

- **Source:** World Bank Open Data API (v2), plus the countries endpoint for region, income level and coordinates.
- **Period:** 2016 onwards. The data snapshot appears to date from around May 2025, which is why 2024 is almost empty.
- **Coverage:** 26 indicators in 7 themes: economic, labour, trade, poverty and inequality, environment, health (13 indicators), and technology. See Appendix A.

3.2 Pipeline

1. Fetch the countries list and flatten the nested fields (region, incomeLevel, lendingType).
2. Loop over indicator groups, paginate through the API (1,000 rows per page), and keep years after 2015.
3. Merge each theme with the country attributes on ISO2 country_id (inner join) and drop unneeded columns.
4. Load the 7 theme tables into Power BI through Power Query Python.Execute.

3.3 Data model and measures

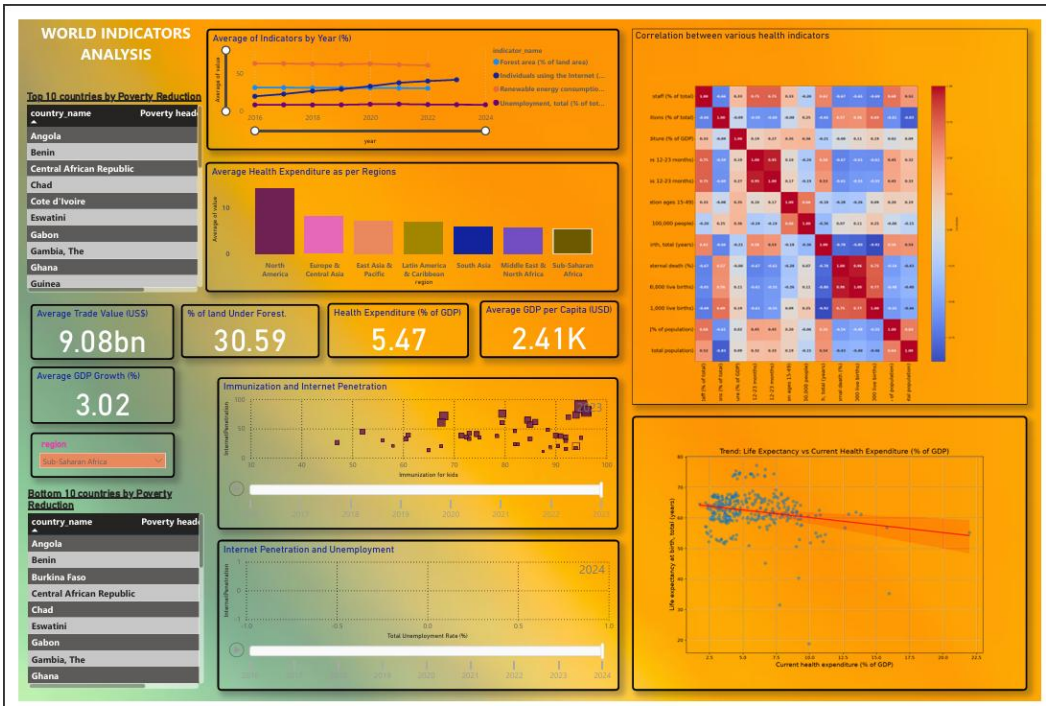
- A dim_table of distinct country_id links to all fact tables (many-to-one relationships).
- Custom measures: Average Trade Value, Poverty Reduction % (start versus end year per country), and scatter measures for GDP per capita, internet penetration, immunization and unemployment.

3.4 Analytical choices

- **Aggregates excluded.** World, income groups and regions would double-count countries.
- **Medians for skewed indicators.** In 2022, mean GDP per capita was $20,520 and the median was $7,606.
- **Spearman correlation and log GDP** for relationships that are monotonic but not linear.
- **Single-year cross-sections** (2021–2023, depending on coverage) so that country sets don't shift between years.

4. The Dashboard

- **Header and KPI cards:** average GDP growth, GDP per capita, forest area, health expenditure and trade value.
- **Charts:** average health expenditure by region (column), indicators by year (line), and two animated bubble scatters (immunization vs internet, unemployment vs internet).
- **Poverty tables:** top 10 and bottom 10 countries by poverty change, with a region slicer.
- **Python visuals:** a regression of life expectancy on health expenditure, and a health-indicator correlation heatmap.



1. Understand the overall economic and social landscape

To get a clear baseline of the countries we are analyzing, we need to understand key average economic and development metrics. Specifically, we want to know:
- What is the average GDP per capita, giving us a sense of individual economic well-being?
- What is the average trade value, showing how active countries are in international markets?
- How much are countries spending on health relative to their GDP, which indicates national health investment priorities?
- What is the average GDP growth rate, revealing overall economic momentum?
- What proportion of the total area in these countries is classified as land, which affects land use, agriculture, and natural resource policies.

These metrics provide foundational context before diving into deeper analyses.

2. Compare health spending patterns across regions
Health expenditure is a key factor in understanding how governments prioritize public health. We need to compare average health spending (as a % of GDP) **across different world regions** to see which regions invest more or less in health relative to their economies. This will help highlight regional disparities and may point to broader differences in health outcomes or priorities.

3. Analyze trends in key socio-economic indicators over time
We want to understand how critical indicators — such as forest area, mobile subscriptions, internet subscriptions, inflation, renewable energy use, and unemployment — have **changed over time**. Are countries making progress on environmental sustainability? How fast is technology adoption growing? Are inflation and unemployment stabilizing or fluctuating? These trend insights will inform whether countries are moving in positive or negative directions in their development pathways.

4. Evaluate the impact of internet access on immunization awareness
One important question is whether **increased internet access** has actually contributed to **better public health outcomes**, particularly in raising awareness about immunization. We want to assess if there's a relationship between higher internet penetration and higher immunization rates, suggesting that digital connectivity may be improving health information access and public engagement.

5. Assess the role of internet access in reducing unemployment
Another important area is the potential economic impact of the internet. Specifically, has expanding internet penetration helped to **reduce unemployment levels**? Understanding this relationship will shed light on whether digital inclusion is translating into better labor market outcomes, possibly by enabling access to job opportunities, skills, or entrepreneurship.

6. Identify underperforming countries in poverty reduction efforts
To identify where development interventions are most urgently needed, we need to pinpoint the **bottom 10 countries** that have shown the **least progress** in reducing poverty over time. This will allow policymakers, NGOs, and donors to focus resources and strategies on the countries struggling the most.

7. Identify top-performing countries in poverty reduction efforts
On the flip side, we also want to learn from success stories. Which are the **top 10 countries** that have made the **most progress** in reducing poverty over the years? Understanding the characteristics and policies of these high-performing countries can provide valuable lessons for others.

8. Examine interrelationships among health indicators
To improve health outcomes, it's crucial to understand how different health indicators **relate to each other**. We want to explore the correlations between factors like health expenditure, life expectancy, immunization rates, child mortality, and disease burden. Identifying strong or weak relationships will help shape integrated health strategies.

9. Investigate the relationship between spending and life expectancy
Lastly, we want to explore whether **increased investment in health expenditure** is actually driving **improvements in life expectancy using trend line**. Is there a clear relationship,This analysis will inform whether boosting spending is an effective lever for improving population health or if other factors need to be addressed alongside it.

year	InternetPenetration	Immunization for kids
2016	51.13	87.21
2017	54.56	87.09
2018	57.33	87.36
2019	60.26	87.56
2020	63.79	85.08
2021	68.62	83.72
2022	70.39	84.65
2023	71.96	85.09
2024	82.16	
Total	**62.18**	**85.97**

5. Data Quality Findings

Issue	Evidence	Consequence
2024 almost empty	Most indicators have 0 countries in 2024	Effective window is 2016–2023
Sparse indicators	Cause-of-death 9% filled, poverty 20%, skilled birth attendance 24%, Gini 25%	Conclusions on these are tentative
Changing country sets	2023 health spending: 21 countries; 2024 internet: 9 countries	Year-to-year averages jump artificially
Aggregates in the data	49 of 266 entities	Distorts every average
Trade reporting gap	World exports exceed imports every year (about 2% in 2022)	Reflects reporting asymmetry

6. Key Insights

6.1 Income decides almost everything

Medians by income group (latest available year):

Indicator	High	Upper-middle	Lower-middle	Low
GDP per capita, US$ (2022)	34,861	7,069	2,343	745
Life expectancy, years (2023)	81.2	74.0	68.6	63.6
Infant mortality per 1,000 (2023)	3.6	11.2	24.5	43.7
Maternal mortality per 100k (2023)	8	42	123	352

Indicator	High	Upper-middle	Lower-middle	Low
Internet users, % (2023)	93.2	80.1	57.8	20.5
Basic water access, % (2022)	100	97.2	86.6	62.4
TB incidence per 100k (2023)	7	46	149	144
Population 65+, % (2023)	17.4	8.1	4.3	2.9

A child in a low-income country faces a 12× higher infant mortality rate and a maternal mortality ratio 44× higher than in a high-income country. Life expectancy differs by 17.6 years.

6.2 Health spending (% of GDP) barely explains life expectancy

- Correlation of health expenditure with life expectancy is weak: Pearson 0.24 across country-years (Spearman 0.37).
- Log GDP per capita correlates 0.84 (Pearson) with life expectancy.
- In the 2021 cross-section, income alone gives $R^2 = 0.70$. Adding health expenditure leaves R^2 at 0.70, with a coefficient of only 0.03 years per percentage point of GDP.
- Low-income countries spend 5.8% of GDP on health, more than lower-middle-income countries (4.9%), yet live 5 years less. A share of a tiny GDP buys little.
- **Over-performers** (life expectancy above what income predicts, 2021): Sri Lanka (+7.4 years), Kyrgyz Republic (+7.8), Thailand (+6.2).
- **Under-performers:** Central African Republic (−19.3), Nigeria (−12.4), Eswatini (−10.6), South Africa (−9.3).
- **Caveat:** per-capita health spending would be a better measure, and this is correlation, not causation.

6.3 The pandemic fingerprint

- **GDP growth** averaged 3.0–3.2% in 2016–2019, fell to **−5.1% in 2020** (median −3.8%), rebounded to **+5.6% in 2021**, and returned to 3.2% by 2023.
- **Life expectancy fell in 84% of countries** between 2019 and 2021 (median −0.9 years).
    - Latin America and the Caribbean fell 1.8 years and the Middle East and North Africa 1.9.
    - Sub-Saharan Africa fell only 0.4, though these are modeled estimates.
    - Bolivia fell 6.4 years, Paraguay 5.6 and Mexico 4.8.
    - By 2023 only 14% of countries were still below their 2019 level.
- **Immunization slipped.** Median DPT coverage went from 93% (2019) to 91% (2022), and 51% of countries were lower in 2022 than in 2019. Low-income countries lost 2.5 points at the median.
- **Infant mortality kept falling** through the pandemic (median 14.05 in 2019 to 13.3 in 2023).
- **Youth unemployment** jumped from a median of 11.8% (2019) to 15.3% (2020), then eased to 12.3% by 2023.

6.4 Digital catch-up

- Median internet use rose from 54% (2016) to 81% (2023).
- Gains between 2016 and 2022 were largest in the middle: lower-middle income +26.1 points, upper-middle +22.4, low income +13.8, high income +9.4. By region, South Asia led with +25.2, and North America gained only +4.8.
- The 40 countries below 20% in 2016 moved from a median of 11.6% to 27.6%.

- Starting level and gain are negatively related (Spearman −0.52), which is the pattern of a catch-up effect.
- Biggest gainers: Mongolia (+59.9 points), Iraq (+58.8), Guyana (+42.7).
- Internet use is a near-perfect development proxy: 0.88 with GDP per capita, 0.84 with water access, −0.83 with infant mortality.
- **It has no relationship with unemployment** (Spearman 0.03), which is exactly what your internet vs unemployment scatter should show.

6.5 Youth unemployment is a regional story

- Youth unemployment is a median **2.3×** the total rate (12.3% vs 5.3% in 2023).
- It is worst in the Middle East and North Africa (22.5% youth against 9.0% total), followed by Latin America (16.8%) and South Asia (15.3%).
- Extremes in 2023: Djibouti (76.3%), South Africa (59.7%), Eswatini (58.2%).
- Unemployment is **uncorrelated with GDP per capita** (Spearman 0.00). Rich and poor countries both have high and low unemployment.
- Sub-Saharan Africa's low reported youth rate (8.5%) likely reflects informal work being counted as employment. This is a hypothesis, not tested here.

6.6 Trade is concentrated

- In 2022, the top 5 exporters accounted for **34.9%** of exports across 178 countries, and the top 10 for **49.3%**.
- China exported $3.72T, the US $3.02T, and Germany $1.91T.
- The US had the largest deficit (−$959B) and China the largest surplus (+$578B). 65% of countries ran a trade deficit.

- Total exports rose from $20.3T (2016) to $30.7T (2022). They fell 9.5% in 2020 and rebounded 25% in 2021.

6.7 Wealth is extremely unequal between countries

- In 2022, the top 10 countries by GDP per capita averaged **238×** the bottom 10.
- Regional medians range from $1,341 (Sub-Saharan Africa) to $78,035 (North America).
- Within countries, inequality is only weakly related to income (Gini vs GDP per capita, Spearman −0.33). Latin America has the highest median Gini (44.8).

6.8 The renewable energy paradox

- Low-income countries get a median **76%** of energy from renewables versus 11.6% in high-income countries (2021), with a correlation of −0.53 with GDP per capita.
- This is most likely traditional biomass in poorer countries, not clean-energy leadership. The indicator measures share, not clean capacity.

6.9 Poverty progress is real but thinly measured

- Only 122 of 217 countries report poverty data, and only 71 report at least two points.
- Of those 71, poverty fell in 43 and rose in 26 (median change −0.8 percentage points).
- 41 of the 71 are in Europe and Central Asia. The Middle East and North Africa and South Asia have one country each, so **regional comparisons are unreliable**.

- Largest declines: Viet Nam (9.2% to 4.3%), Uzbekistan (17.0% to 11.0%), Georgia (22.0% to 15.6%).
- Largest rises: Peru (20.7% to 27.5%), Argentina (30.3% to 39.2%), Kyrgyz Republic (25.4% to 33.3%).

6.10 Forests are roughly flat

- Median change in forest cover from 2016 to 2022 is 0.0 points. 36% of countries lost more than 0.05 points and 27% gained.
- Biggest losses: Cambodia (−5.3 points), Nicaragua (−5.0), Paraguay (−4.4).
- The Middle East and North Africa median is just 1.0% forest cover.

7. Dashboard Review and Corrections

Issue	Effect	Resolution
Aggregates included in all averages	Average trade value 6.7× too high ($9.29×10^{11} vs $1.39×10^{11}); GDP per capita card understated (17,542 vs 19,029)	Page filter excluding Aggregates
Trade measure searched for “Import…” instead of “Imports…”	Imports never matched, so the card showed exports only	Card rebuilt with a filter on both indicators
Poverty “Top 10” sorted the wrong way	Showed countries where poverty rose most	Table titles swapped, column renamed to “Poverty headcount change (%)”

Issue	Effect	Resolution
Scatter charts used year as category	About 9 global-average bubbles instead of countries	Category changed to country name
“Youth unemployment” measure used total unemployment	Mislabeled axis	Relabeled (youth indicator is available in the data)
Line chart mixed mobile subscriptions (per 100 people) with percentage series	Scale dominated by one line	Mobile series removed
Local CSV combined into two queries	2,394 duplicate rows, and refresh fails on other machines	Remove the CSV from the combine step
Regression visual used pivot()	Breaks on duplicate rows	Switched to pivot_table()

8. Limitations

- Many indicators are modeled estimates (life expectancy, mortality, unemployment).
- Missing data is heavy for poverty, Gini and cause of death, and the effective window is 2016–2023.
- Averages are unweighted by population, so small countries count equally with large ones.
- All relationships here are correlations, not causal effects.
- Some residual findings (for example Syria’s life expectancy) reflect modeled data and should be read cautiously.

9. Conclusions and Recommendations

- Development outcomes track income closely, but health spending as a share of GDP is a weak lever on its own.
- The pandemic reversed several years of progress in life expectancy and immunization, and most countries have since recovered.
- Digital access is converging fastest in middle-income countries.
- Dashboards should always exclude aggregates, state the number of countries behind each average, and use medians for skewed indicators.
- Poverty conclusions need more data before any regional claim is made.

10. Future Work

- Add per-capita health spending (SH.XPD.CHEX.PC.CD) and test its effect on life expectancy.
- Use population-weighted averages and a panel regression with country fixed effects.
- Cluster countries (for example k-means) on development indicators.
- Refresh to pull 2024 and later data, and move to scheduled refresh through a gateway.